

1. (15%) Protocol layers. (Introduction)

- What are the five protocol layers, from top to bottom, in the Internet? (5%)
- Describe the main functionalities of each of the five protocol layers. (10%)

2. (15%) Packet Loss and delays. (Introduction)

As a packet travels from one node to the subsequent node along a network path, the packet suffers from different types of delay at each node. The most important of these delays are: (i) nodal processing delay (d_{proc}), (ii) queuing delay (d_{queue}), (iii) transmission delay (d_{trans}), and (iv) propagation delay (d_{prop}).

- Please describe each of them briefly. (8%, 2 points each)
- What are the most critical factor (that is, the most lengthy delay might be caused) among the above four delay factors? And why? (3%)
- Suppose with a network cable, the network signal travels at speed of 3×10^7 meters per second. What are the propagation delay (in second) to send a network packet over the network cable connecting between two nodes with a distance of 40 000 km (roughly the same distance to circle around the

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3. (15%) internet check-sum

(a) Given a sequence of 9 bytes as [0xF1, 0x01, 0xF2, 0x02, 0xF3, 0x03, 0xF4, 0x04, 0xF9]. What is the checksum number of this 9-byte network segment, according to the internet checksum algorithm? (5%) (hint: check sum is 16-bit)

(b) Continue from (a), suppose now we insert two zero bytes (that is, 0x00 0x00) into before the third byte of above sequence and form a new sequence of 11 bytes as [0xF1, 0x01, 0x00, 0x00, 0xF2, 0x02, 0xF3, 0x03, 0xF4, 0x04, 0xF9]. Explain that the checksum value of the new 11-byte sequence should be the same as the 9-byte sequence shown in (a) (5%)

(c) Continue from (b), suppose the checksum value of the 9-byte sequence shown in (a) is 0xWX 0xYZ. Now similar to (b) we insert this two-byte sequence (0xWX 0xYZ) into before the third byte of 9-byte sequence and form a new sequence of 11 bytes as [0xF1, 0x01, 0xWX, 0xYZ, 0xF2, 0x02, 0xF3, 0x03, 0xF4, 0x04, 0xF9]. Explain that the checksum value of the new 11-byte sequence should be zero. (5%)

4. (15%) reliable data transfer. In the rdt 3.0 (reliable data transfer version 3.0), we set the window size = 1, and associate each packet sent with label 0 or 1. The binary label (0 or 1) scheme will help us to solve the packet loss and packet duplication problem.
- (a). Figure 1 is the FSM (finite state machine) for rdt 3.0 (sender part). Please fill in the nine missing statements labeled with "?1" through "?9". (9%)
 - (b). Please give a short example to explain why the binary label scheme can solve packet duplication problem. (3%)
 - (c). Let the notation P^s , $s=\{0,1\}$ represents the receiver receives a packet P with label of s . For example $A^0B^1C^0$ represents the receiver receives three packets in order as A (with label 0), B (with label 1), and C (with label 0). Now, suppose a receiver receives a sequence of the following 15 packets: $A^0A^1A^1A^1B^0A^1A^1B^0B^0A^1A^1A^1C^0C^1C^0$. What is the right packet sequence the sender sends? (3%)

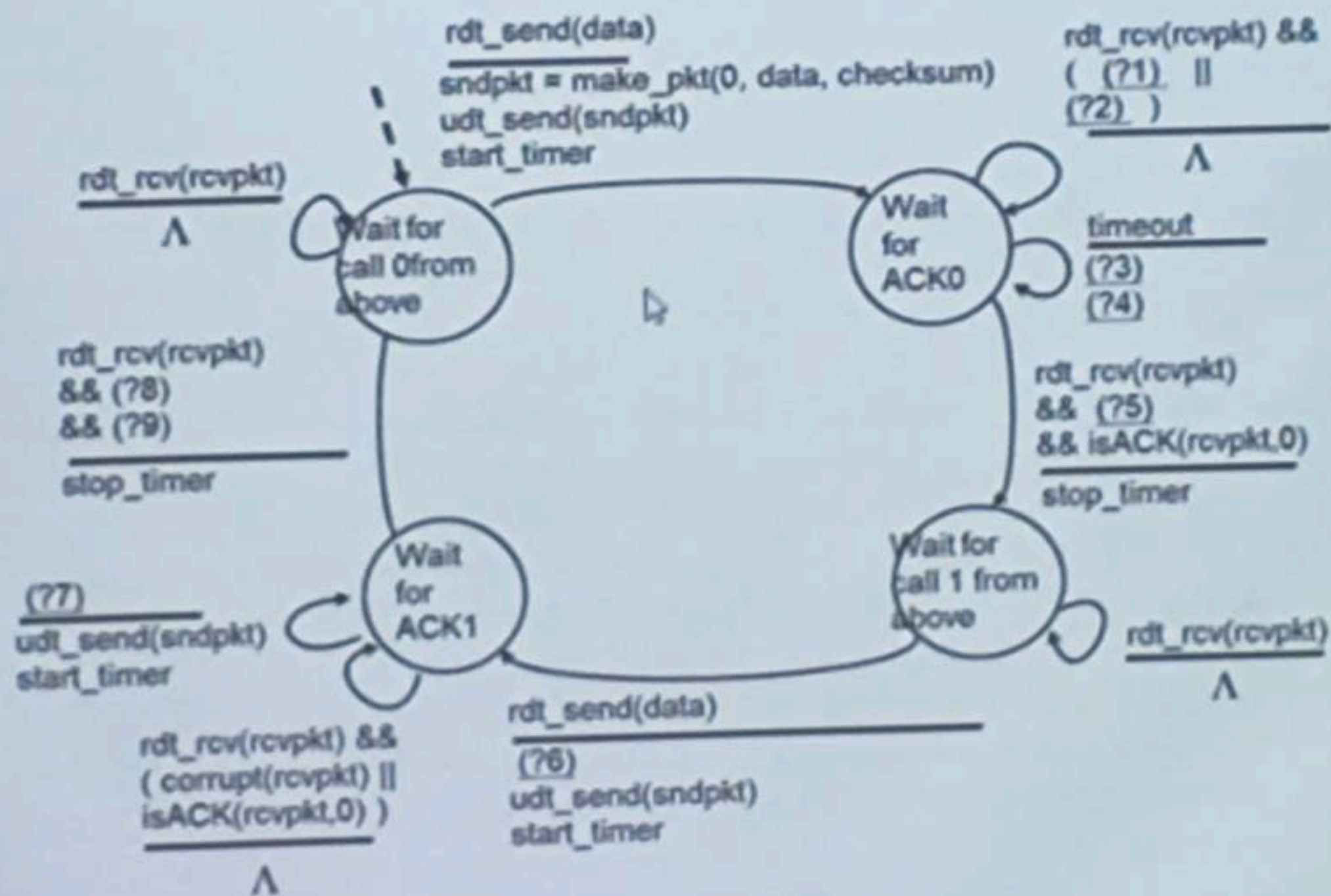


Figure 1. FSM for rdt 3.0

5. (18%) selective repeat (SR) vs. go-back N (GBN).
- (a). (6%) ACK mechanism: Individual ACK vs. Cumulative ACK. 1. Please Explain the difference between these two ACK mechanism. (2%); 2. What is the benefit of Individual ACK over Cumulative ACK? Why? (2%); 3. What is the benefit of Cumulative ACK over Individual CK? Why? (2%)

- (b). (3%) SR vs. GBN in ACK mechanism: What is the differences in ACK mechanism between the two? For example, if the sender receives a ACK number 5, what does it mean with GBN scheme? and, what does it mean with SR scheme? (hint: Individual ACK vs. Cumulative ACK)
- (c). (3%) SR vs. GBN in out-of-order packet: What are the difference on receiving an out-of-order packet between the two? For example, if now the next packet expected to receive is with sequence number of 5 but we get a packet with sequence number of 8, how the receiver side deals with this out-of-order packet in GBN and SR respectively?
- (d). (3%) SR vs. GBN in Timer: How many timer is required for the sender with GBN? and how many with SR?
- (e). (3%) What actions would be taken at the sender side in the case of GBN and SR respectively, if a packet time-out event happens?

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● Go-back-N: sender windows



Figure 2. transport layer control: GBN and Selective Repeat mechanisms.

6. (14%) (from NTU EE Prof. Polly Huang, 2019 Midterm Exam)

(Overview) ping is a commonly used network testing tool. What it does is to send a probing message periodically to a remote host. The remote host, upon receiving the probing message, sends back a reply message. From the reply

message, the probing host calculates the round-trip time (RTT) between the two

message, the probing host calculates the round-trip time (RTT) between the two. By tracking the number of probing messages sent and the reply messages received, the probing host calculates the loss rate as well.

Below is the result of a ping test to `www.eurocom.fr` from Polly's office desktop PC. The command is interrupted manually at the 196th test. One can see that the RTT for the 1st probe is 409.086 ms. One can also see in the ping statistics that the average RTT is 345.159 ms.

```
pollyss-iMac:~ pollyoffice$ ping www.eurocom.fr  
PING eurocom.fr (217.70.191.35): 56 data bytes
```


PING eurocom.fr (217.70.191.35): 56 data bytes

64 bytes from 217.70.191.35: icmp_seq=0 ttl=40 time=409.086 ms

64 bytes from 217.70.191.35: icmp_seq=1 ttl=40 time=428.191 ms

...

64 bytes from 217.70.191.35: icmp_seq=195 ttl=40 time=315.882 ms

64 bytes from 217.70.191.35: icmp_seq=196 ttl=40 time=328.620 ms

^C

--- eurocom.fr ping statistics ---

197 packets transmitted, 195 packets received, 1.0% packet loss

round-trip min/avg/max/stddev = 281.737/345.159/468.015/47.067 ms

pollyss-iMac:~ pollyoffice\$

- (1) Find the (a) packet loss rate, the (b) minimum, (c) maximum, and (d) standard deviation of the RTT. (4%)
- (2) The RTT observed by the 2nd probe is longer than the 1st probe. State at least 1 reason that might have contributed to the difference. (2%)

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^C

--- eurocom.fr ping statistics ---

197 packets transmitted, 195 packets received, 1.0% packet loss

round-trip min/avg/max/stddev = 281.737/345.159/468.015/47.067 ms

pollyss-iMac:~ pollyoffice\$

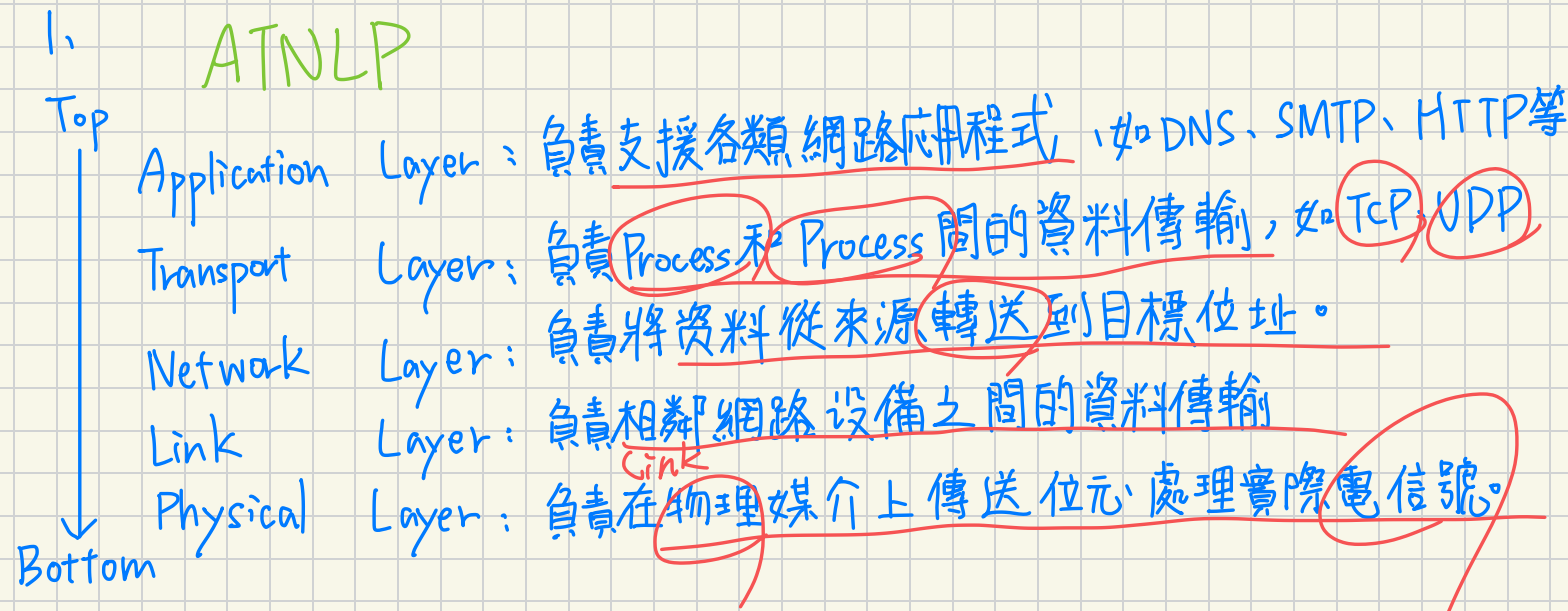
- (1) Find the (a) packet loss rate, the (b) minimum, (c) maximum, and (d) standard deviation of the RTT. (4%)
- (2) The RTT observed by the 2nd probe is longer than the 1st probe. State at least 1 reason that might have contributed to the difference. (2%)
- (3) Do you think ^I217.70.191.35 is in Taiwan? Why or why not? (2%)
- (4) If you need to estimate the average RTT from your host to another host, would you prefer to use traceroute or ping? And why? (3%)
- (5) If you need to find out which router on the way from your host to another host contributes the most to the RTT, would you prefer to use traceroute or ping? And why? (3%)

(4) If you need to estimate the average RTT from your host to another host, would you prefer to use traceroute or ping? And why? (3%)

(5) If you need to find out which router on the way from your host to another host contributes the most to the RTT, would you prefer to use traceroute or ping? And why? (3%)

7. (8%) course feedback.

To improve course quality, please give your comments if any. Any comments are welcome. No matter what you write, you'll get full score of this question. If you don't know what to write, you can simply write down some short words, like "OK" or "no comment". Thanks!!



2. (a) d_{proc} = 當封包進入 router 後做基本處理的延遲, 如檢查 bit error, header 等行為。
- d_{queue} = 若同時有許多封包要走相同的 output link 出去, 後抵達的封包就必須排隊等候。
- d_{trans} = 將整個封包推上金線路所需的時間。
- d_{prop} = 封包在金線路上從起點到終點所需的時間。

(c) $s = 3 \times 10^7 \text{ m/s}$

$d = 40000 \text{ km} = 40000000 \text{ m} = 4 \times 10^7$

$\frac{4 \times 10^7}{3 \times 10^7} = \frac{4}{3} \approx 1.33 \text{ s}$

(b) queueing delay (d_{queue})

queueing delay 是 4 種延遲中不確定性最高、最難預測的。

其他三種延遲相對固定, 由硬體、網路頻寬、距離等決定, 但 queueing delay 取決於當下網路環境的擁塞情況, 在尖峰期有可能從 0 暴增至數秒。

3. Chapter 3 P16 UDP

(a) 給定一 9 byte 序列, 求 CheckSum?

How to Calculate CheckSum?

- 將 bytes 先兩一組湊一對 (若 byte 數為奇數, 多補 0)
- 相加 (若有進位, 要加回最低位)
- 取補 ($0xFFFF - X$)

$0xF1 \ 0x01 \rightarrow 0xF101$	$\begin{array}{r} F101 \\ + F202 \\ \hline 1E303 \end{array}$	$\begin{array}{r} E304 \\ + F303 \\ \hline 1D607 \end{array}$	$\begin{array}{r} D608 \\ + F404 \\ \hline 1CA0C \end{array}$	$\begin{array}{r} CA0D \\ + F900 \\ \hline 1C30D \end{array}$	$\begin{array}{r} FFFF \\ - C30E \\ \hline 3CF1 \end{array}$
$0xF2 \ 0x02 \rightarrow 0xF202$	$\begin{array}{r} E303 \\ + 1 \\ \hline E304 \end{array}$	$\begin{array}{r} D607 \\ + 1 \\ \hline D608 \end{array}$	$\begin{array}{r} CA0C \\ + 1 \\ \hline CA0D \end{array}$	$\begin{array}{r} C30D \\ + 1 \\ \hline C30E \end{array}$	Ans: $0x3CF1$
$0xF3 \ 0x03 \rightarrow 0xF303$					
$0xF4 \ 0x04 \rightarrow 0xF404$					
$0xF9 \ 0x00 \rightarrow 0xF900$					

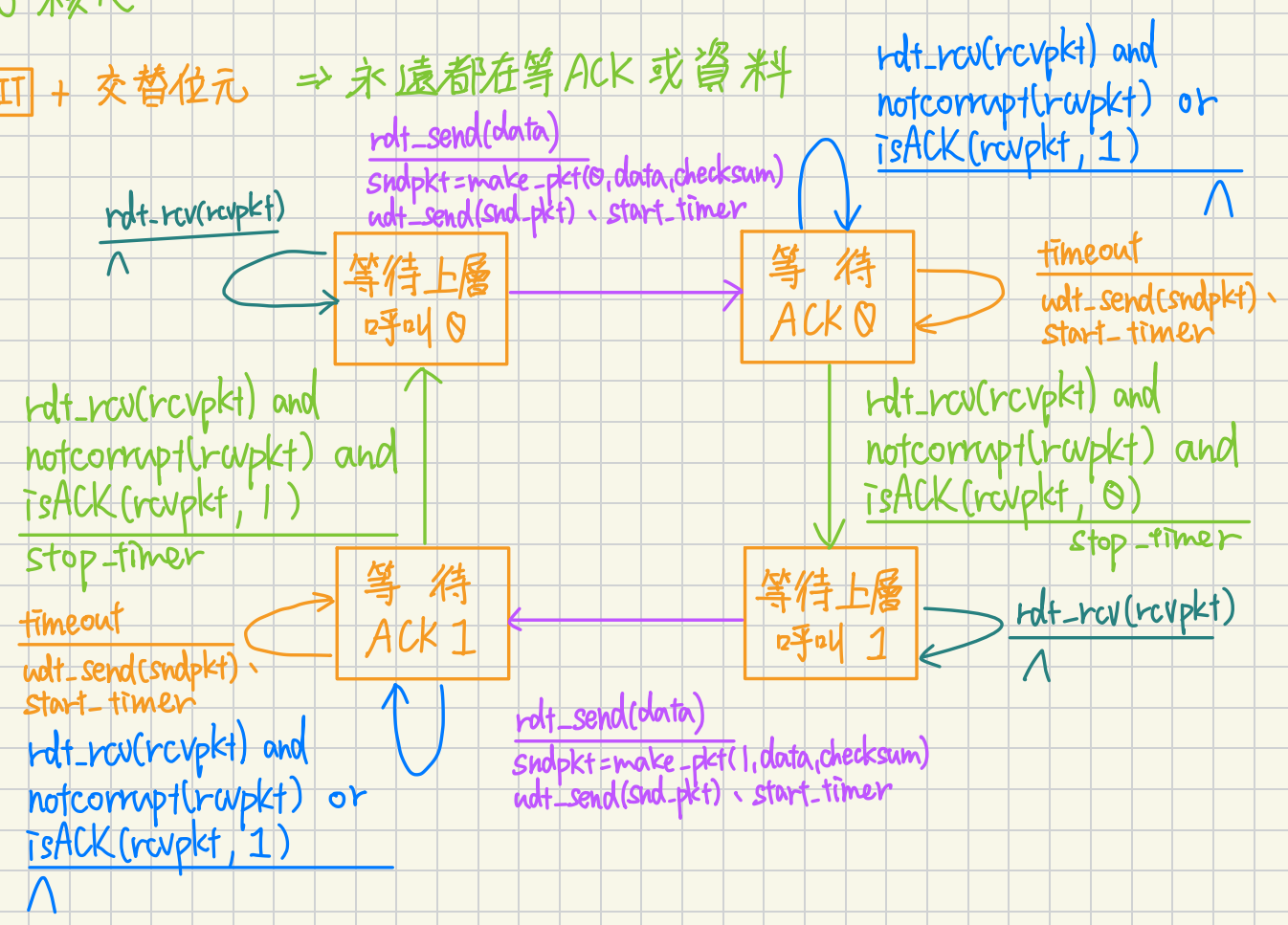
(b) 插入的新 byte 皆為 $0x00$, 組合後會變成值為 0 的 16-bit 字組, 在加總與取補時皆不會對原始值有任何貢獻, 自然 CheckSum 也就不變。

(c) 根據定義, 原始資料之總和 + CheckSum = $0xFFFF$ (全為 1), 這導致了若在計算原資料總和時, 若將 CheckSum 一起加入計算, 其計算結果必為 $0xFFFF$, 經取補完就變成 0 了。

4 Chapter 3 P28 可靠資料傳輸原理

RDT 3.0 核心

STOP & WAIT + 交替位元 $\Rightarrow$ 永遠都在等 ACK 或資料



等待 ACK 時的 3 狀態:

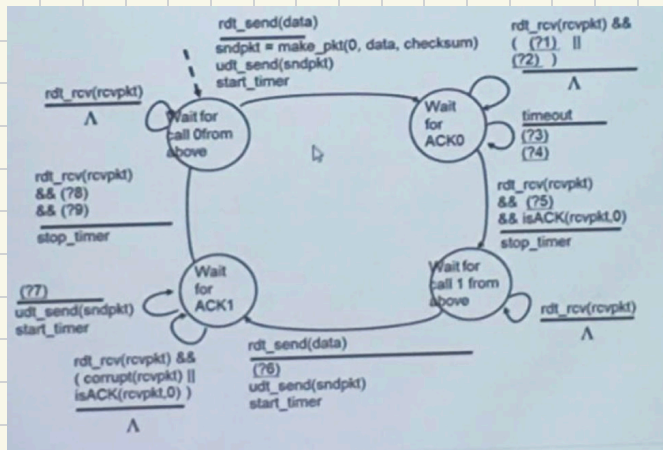
- 下一步: 收到封包 且 封包沒壞 且 編號正確
 $rdt_rcv(rcvpkt)$ and $not\ corrupt(rcvpkt)$ and $isACK(rcvpkt, 0/1)$
 $\Rightarrow$ 停止計時 (stop-timer)
- 封包壞了、序號不對: 收到封包 但 封包壞了 或 編號錯了
 $rdt_rcv(rcvpkt)$ and $corrupt(rcvpkt)$ or $isACK(rcvpkt, 0/1)$
 $\Rightarrow$ 什麼都不做 (A), 一直等, 直到 Timeout
- 超時: 計時器超時 (timeout)
 $\Rightarrow$ 重傳之前的封包 並 重啟計時器
 $wdt_send(sndpkt)$ and $start_timer$

等待上層時 2 狀態:

- 上層來資料了: 當上層的 Application 有資料要傳送 ($rdt_send(data)$)
 $\Rightarrow$ 封裝封包計算校驗和、發資料、開計時
 $sndpkt = make_pkt(0/1, data, checksum)$ $wdt_send(sndpkt)$ $start_timer$
- 收到來自網路的封包 $rdt_rcv(rcvpkt)$
 在等待資料時突然收到封包 $\rightarrow$ 過去遲到、重覆的封包
 $\Rightarrow$ 什麼都不做 (A), 忽略繼續等。

4.

(a)



- (1) corrupt(rcvpkt)
- (2) isACK(rcvpkt, l)
- (3) udt_send(sndpkt)
- (4) start_timer
- (5) notcorrupt(rcvpkt)
- (6) sndpkt = make_pkt(1, data, checksum)
- (7) timeout
- (8) notcorrupt(rcvpkt)
- (9) isACK(rcvpkt, l)

(b) 假设 Sender 送出 packet A^0 , receiver 正确收到并回传 ACK0, 但 ACK0 在传输时丢失, sender 等候超时后重传 A^0 , 但此时 receiver 正在等待封包 1, 发现封包是 0, 判定是重复封包捨棄, 并再次发送 ACK0 给 sender

(c) <key> $0 \rightarrow 1$

$A^0 A^1 A^1 A^1 B^0 A^1 A^1 B^0 B^0 A^1 A^1 A^1 C^0 C^1 C^0$
 $\checkmark \checkmark \times \times \checkmark \checkmark \times \checkmark \times \checkmark \times \times \checkmark \checkmark \checkmark$
 $A^0 A^1 B^0 A^1 B^0 A^1 C^0 C^1 C^0$

5. Chapter 3 P.34 GBN: 一个出包, 全部重传 / SR: 谁出包, 谁重传

Go-Back-N (GBN) — 「出错就全部重来」
 傳送端可以連續送出最多 N 個未確認的封包, 但只有一個出包:
 傳送端送出: 1 2 3 4 5 6 7
 封包 3 遺失!
 GBN 做法: 直接重傳 3, 4, 5, 6, 7 (全部重傳)
 • 接收端不暫存亂序封包, 直接丟掉
 • 就像「出了差錯, 從那裡全部重來」

Selective Repeat (SR) — 「只補傳遺失的」
 一樣可以連續送 N 個, 但出錯時:
 傳送端送出: 1 2 3 4 5 6 7
 封包 3 遺失!
 SR 做法: 只重傳封包 3, 4, 5, 6, 7 已暫存不需重傳
 • 接收端會暫存亂序封包, 等缺少的補齊再交給上層
 • 就像「哪個丟了補哪個」

	GBN	SR
出錯時重傳	全部重傳	只傳遺失的
接收端 buffer	✗ 不需要	✓ 需要
ACK 類型	Cumulative (累積)	Individual (個別)
Timer 數量	1 個	N 個
精度效率	較差 (重傳太多)	較好
實作複雜度	簡單	複雜

Go-Back-N:

累積確認 Cumulative ACK

Sender: 只要 window 還有空間, 就一直傳 Packet, 但只幫最老的 Packet 設 Timer。

Receiver: 只收正確的封包, 錯的丟棄。

→ 當 Packet 5 發生問題時, 就算 6、7、8 Packet 平安抵達也會被 Receiver 仍掉。
 待 5 的 timer 逾時後, 5、6、7、8 一起重送。

Selective Repeat:

個別確認 Individual ACK

Sender: 每個封包都有個自己的 Timer

Receiver: 若封包 5 掉了, 但 6、7 抵達, 會將 6、7 先放 Buffer, 並回傳 ACK 6、ACK 7

→ 當 5 號逾時, Sender 只會重傳 5 號。Receiver 收到 5 號後, 會將 Buffer 的 6、7 一起交給 Application

(a)

1.

Individual ACK: 每個封包單獨確認, Receiver 會為每個封包都回傳 ACK。

Cumulative ACK: 確認該序號以前的所有封包都有收到, 例如 ACK 5 代表 5 號以前的封包都已收到。

2. 效率高、減少頻寬浪費。

當某個封包遺失時, Sender 只需重傳該特定封包, 不需要重傳之後已成功抵達的封包

3. 具備較好的抗丟包能力

若中間某一個 ACK 訊息遺失也沒關係, 只要後續有更大的 ACK 抵達就可以涵蓋之間的確認

(b)

SR: 僅代表 5 號封包收到, 不確定 1~4 是否收到

GBN: ACK 5 代表 5 號含以前的封包全都收到了。

(c)

SR: 先將 8 放到 Buffer 中, 回傳 ACK 8, 繼續等待 5

GBN: 直接捨棄封包 8, 並回傳最後一個正確收到封包的 ACK (ACK 4)

(d)

SR: 每個已發送但未確認的封包都各自需要一 timer

GBN: 1 個 timer, GBN 只追蹤最舊且尚未確認的封包。

(e)

SR: 哪個封包 timeout 重傳哪個

GBN: 重傳 window 中所有已傳送但未 ACK 的封包 (send_base ~ nextseqnum-1)

6、

1、

- (a) 1.0%
- (b) 281.737 ms
- (c) 468.015 ms
- (d) 47.067 ms

2、

第2个Packet Queueing Delay 較高、路由路徑不同、遠端主機的負載高低

3、

不在台灣。

因為平均RTT為 345.159ms, 最小RTT也有 281.737ms, 台灣到台灣的RTT通常落在 10ms 左右, 高RTT代表距離遙遠, 加上域名的 .fr 是法國網域, 因此推斷 271.70.191.35 在歐洲。

4、

ping

ping 可以發送大量探測封包, 取平均值更精確。

traceroute 只會顯示每個路由節點的RTT, 無法直接得到端對端的平均RTT。

5、

traceroute

traceroute 會顯示每個路由節點與其對應的RTT, 比較節點之間的RTT差值, 就能知道哪一段造成最大延遲。

O: same as the 2023
O: need to do

National Taiwan Ocean University
Department of Computer Science

Computer Networks — 2025 Midterm Examination

Scope: Chapter 1 ~ Chapter 3

Instructions: This is a closed-book exam. Answer all questions. Write clearly and justify your answers where required. Total: 100 points.

1. (15 points) Internet Protocol Stack

Describe the five-layer Internet protocol stack. For each layer, explain its main function and give at least one example of a protocol used at that layer.

(Hint: The five layers are — Application, Transport, Network, Link, Physical.)

2. (15 points) Packet Delay

(a) Briefly define each of the four types of packet delay that a packet experiences as it travels through a router/node:

1. Nodal Processing Delay: _____

2. Queuing Delay: _____

3. Transmission Delay: _____

4. Propagation Delay: _____

(b) Among the four types, which one is most likely to cause extremely long delays under heavy network load, and why?

(c) A packet has a size of $L = 40,000$ bits. It is transmitted over a link with a bandwidth of $R = 3$ Mbps. What is the transmission delay?

3. (25 points) Circuit Switching vs. Packet Switching

(a) Explain how **circuit switching** works. What are its main characteristics, and what are the drawbacks compared to packet switching?

(b) Explain how **packet switching** works. What are its advantages over circuit switching?

Suppose users share a **1 Mbps** link. Each user requires **100 kbps** when actively transmitting, but each user transmits only **10%** of the time.

(c) If **circuit switching** is used, how many users can be supported simultaneously?

(d) Now suppose **packet switching** is used with **40 users**. Using the binomial distribution, write an expression for the probability that exactly n users are transmitting simultaneously at any given time.

(e) Find the probability that there are **11 or more** users transmitting simultaneously. (You may leave your answer in summation form.)

(f) In packet switching, a router uses **store-and-forward** transmission. Explain what this means — why must the router wait until the entire packet arrives before forwarding it?

(g) What problem might arise if a router forwarded each bit immediately upon arrival, without waiting for the complete packet?

4. (10 points) DNS Resource Records

For each of the following descriptions, identify the type of DNS resource record it corresponds to. Choose from: **A, NS, MX, CNAME**.

#	Description	Record Type
(a)	Maps a mail server hostname for a domain (e.g., directs email for foo.com to mail.foo.com)_____	
(b)	Provides an alias for a canonical hostname (e.g., www.foo.com → relay1.bar.foo.com)_____	
(c)	Maps a hostname directly to its IPv4 address	_____
(d)	Identifies the authoritative name server for a domain	_____
(e)	The record returned by an authoritative DNS server that provides the IP address of a <u>host</u> _____	
(f)	The record type stored in a root or TLD DNS server pointing to the next DNS server in <u>the hierarchy</u>	

5. (10 points) Network Performance Metrics for Different Applications

For each of the following network applications, discuss which performance metrics (choose from: **throughput, delay, packet loss, security**) are most critical. Give a brief explanation for each choice.

(a) An online banking / financial transaction system:

(b) A short-video streaming platform (such as TikTok / Reels):

(c) The DNS system (especially when it becomes a target of DDoS attacks):

6. (13 points) Reliable Data Transfer (RDT 3.0)

(a) (9 pts) The figure below shows a simplified RDT 3.0 sender finite-state machine with several labels missing. Fill in the blanks (labeled 1–9) with the correct actions or conditions from the list below:

Available terms: corrupt(rcvpkt), isACK(rcvpkt,0), isACK(rcvpkt,1), notCorrupt(rcvpkt), udt_send(sndpkt), start_timer, stop_timer, timeout, sndpkt = make_pkt(0,data,checksum), sndpkt = make_pkt(1,data,checksum)

Blank [1]: Condition to stay in 'Wait for ACK 0' state when a bad/wrong ACK arrives

Answer: _____

Blank [2]: Condition: correct ACK for packet 0 received

Answer: _____

Blank [3]: Action: resend the current packet

Answer: _____

Blank [4]: Action: start the retransmission timer

Answer: _____

Blank [5]: Condition: packet received is not corrupted

Answer: _____

Blank [6]: Action: create packet 1 with data and checksum

Answer: _____

Blank [7]: Event that triggers retransmission (timer expires)

Answer: _____

Blank [8]: Condition: packet received without corruption (in 'Wait for ACK 1')

Answer: _____

Blank [9]: Condition: correct ACK for packet 1 received

Answer: _____

(b) (4 pts) Trace through the following scenario using RDT 3.0. Describe what happens at each step:

Step 1: Sender transmits packet P0. Receiver gets P0 correctly and sends ACK0.

Step 2: Sender transmits packet P1. The ACK1 from receiver is delayed (not yet received by sender).

Step 3: Sender's timer expires. Sender retransmits P1. Receiver now receives two copies of P1.

Step 4: What does the receiver do upon receiving the duplicate P1? What does it reply?

6.

6. (7 points) Pipelined Protocols — Go-Back-N / Selective Repeat

Consider a scenario where three senders A, B, C each need to send data frames. Suppose the window size is 2 and sequence numbers cycle (0, 1). Frames from sender A are labeled A0, A1; from B are B0, B1; from C are C0, C1.

(a) Trace the correct frame-sending order in the sequence below assuming Go-Back-N is used, and some frames are lost and must be retransmitted. Fill in the blank frames in the sequence:

A0, A1, B0, A1, B____, A1, C0, C1, C0

(b) Explain why A1 appears multiple times in the sequence above. Which protocol behavior does this illustrate?

7.

7. (5 points) TCP Basics

(a) When you open a browser and type a URL (e.g., <https://tronclass.ntou.edu.tw>), which application-layer protocol is used to retrieve the web page?

(b) What is the minimum size (in bytes) of a TCP segment header?

(c) In a TCP segment, the **Acknowledgement Number** field is set to 1. What does this value indicate?

(d) What does it mean when the **Receive Window** field in a TCP segment is set to 0? What should the sender do?

(e) How can a TCP receiver determine where the TCP header ends and the data payload begins in a received segment? Which field in the TCP header provides this information?

3.

(a)

circuit switching 類似傳統的電話系統，在開始資料傳輸前需事先建立連線，使用期間頻寬完全由使用者獨佔，雖然較為穩定，不過相較 packet switching 他無法有效率的利用頻寬資源，即使沒在傳輸，頻寬依然會被佔用。

(b)

packet switching 會將資料分割成一塊塊的封包，每個封包用完整的頻寬傳輸，所有使用者共享頻寬，因此相比起 circuit switching 能較高效的利用有限的頻寬資源。

(f)

資料要傳輸到下一個鏈路前，必須先接收到完整的封包。若不接收到完整的封包，就無法進行錯誤檢測。

(g)

若抵達即轉發，表示沒有做任何錯誤檢測，可能會導致損壞的封包佔用頻寬資源。

4.

(a) MX

(b) CNAME

(c) A

(d) NS

(e) A

(f) NS

7.

(a) HTTP

(b) 一行 32bit (4 bytes)，最少 5 行

$$4 \times 5 = 20$$

(c) receiver 期望收到序號為 1 的資料

(d) Receiver 的 buffer 已滿，無法再接收任何資料

(e) 檢查 TCP Header 中的 header len 欄位